R Code

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################################################################
# PRACTICAL 6: AUTOREGRESSIVE MOVING AVERAGE (ARMA) MODELS
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# Objective:
# To fit ARMA models, estimate their coefficients,
# obtain the fitted equation, and examine residuals.

# ============================================================
# 1. INTRODUCTION TO THE DATASET
# ============================================================

data(WWWusage)

# Display the dataset
WWWusage

# ============================================================
# 2. BASIC INFORMATION ABOUT THE DATA
# ============================================================

class(WWWusage)
length(WWWusage)
start(WWWusage)
end(WWWusage)
frequency(WWWusage)

# ============================================================
# 3. PLOT THE ORIGINAL TIME SERIES
# ============================================================

plot(WWWusage,
     main = "Internet Usage Time Series",
     xlab = "Time",
     ylab = "Internet Usage")

# ============================================================
# 4. ACF OF THE ORIGINAL SERIES
# ============================================================

acf(WWWusage, main = "ACF of WWWusage")

# ============================================================
# 5. PACF OF THE ORIGINAL SERIES
# ============================================================

pacf(WWWusage, main = "PACF of WWWusage")

# ============================================================
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# 6. FIRST DIFFERENCE OF THE SERIES
# ============================================================

dWWWusage <- diff(WWWusage)

# Display differenced data
dWWWusage

# Plot differenced series
plot(dWWWusage,
     main = "First Difference of WWWusage",
     xlab = "Time",
     ylab = "Change in Internet Usage")

# ============================================================
# 7. ACF AND PACF OF DIFFERENCED SERIES
# ============================================================

acf(dWWWusage,
    main = "ACF of Differenced WWWusage")

pacf(dWWWusage,
     main = "PACF of Differenced WWWusage")

# ============================================================
# 8. FIT ARMA(1,1) MODEL
# ============================================================

# The differenced series is modelled using ARMA(1,1).
# order = c(p, d, q)
# p = AR order, d = differencing within arima(),
# q = MA order.
#
# Since we have already differenced the data,
# d = 0 in the model.

arma11 <- arima(dWWWusage,
                order = c(1, 0, 1),
                include.mean = TRUE)

# Display model output
arma11

# ============================================================
# 9. EXTRACT ESTIMATED COEFFICIENTS
# ============================================================

# Display all estimated coefficients
arma11$coef

# Extract AR(1) coefficient
phi1 <- unname(arma11$coef["ar1"])

# Extract MA(1) coefficient
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theta1 <- unname(arma11$coef["ma1"])

# Extract estimated mean
mu <- unname(arma11$coef["intercept"])

# Display individual estimates
cat("Estimated mean =", round(mu, 4), "\n")
cat("Estimated AR(1) coefficient =", round(phi1, 4), "\n")
cat("Estimated MA(1) coefficient =", round(theta1, 4), "\n")

# ============================================================
# 10. DISPLAY THE FINAL ARMA(1,1) EQUATION
# ============================================================

cat("\nFitted ARMA(1,1) equation:\n")

cat("Y(t) =",
    round(mu, 4),
    ifelse(phi1 >= 0, " + ", " - "),
    abs(round(phi1, 4)), "Y(t-1)",
    " + epsilon(t)",
    ifelse(theta1 >= 0, " + ", " - "),
    abs(round(theta1, 4)), "epsilon(t-1)\n",
    sep = "")

# ============================================================
# 11. ESTIMATED INNOVATION VARIANCE
# ============================================================

cat("\nEstimated innovation variance:\n")
print(arma11$sigma2)

# ============================================================
# 12. FIT ARMA(2,1) MODEL
# ============================================================

arma21 <- arima(dWWWusage,
                order = c(2, 0, 1),
                include.mean = TRUE)

# Display model output
arma21

# Display estimated coefficients
arma21$coef

# ============================================================
# 13. COMPARE MODELS USING AIC
# ============================================================

cat("\nAIC of ARMA(1,1):", AIC(arma11), "\n")
cat("AIC of ARMA(2,1):", AIC(arma21), "\n")

# Smaller AIC is preferred among candidate models,
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# provided the model diagnostics are satisfactory.

# ============================================================
# 14. RESIDUAL DIAGNOSTICS
# ============================================================

# Diagnostic plots
tsdiag(arma11)

# Plot residuals
plot(residuals(arma11),
     type = "l",
     main = "Residuals of ARMA(1,1)",
     xlab = "Time",
     ylab = "Residuals")

# ACF of residuals
acf(residuals(arma11),
    main = "ACF of ARMA(1,1) Residuals")

# Ljung-Box test for residual autocorrelation
Box.test(residuals(arma11),
         lag = 12,
         type = "Ljung-Box",
         fitdf = 2)
```